SYSTEMIC RISK AS GEOMETRY: BLIND-SPOT DECOMPOSITION, ADAPTIVE FRICTION, AND APPLICATIONS TO FINANCE, ENERGY, AND NAVIER–STOKES REGULARITY*

ODEYEMI OLUSEGUN†

Abstract. We introduce a geometric framework for analysing instability in coupled dynamical systems. N interacting agents evolve under a potential Φ whose pairwise term encodes contagion; a *Blind-Spot Decomposition Theory* (BSDT) energy functional E_{BS}, calibrated on normal-period data, provides an equivalent Lyapunov function (Theorem C: $c_1\|\operatorname{grad}\Phi\| \leq \|\operatorname{grad} E_{\mathrm{BS}}\| \leq c_2\|\operatorname{grad}\Phi\|$). A *critical manifold* $\mathcal{C}^*$ separates a stable from an unstable regime; adaptive friction $\gamma^*(X) = \alpha/\lambda_{\max}(D^2\Phi_{\mathrm{pair}})$ achieves marginal stability where every fixed coupling fails. The framework is validated on U.S. banking data (FDIC SDI, 1990–2024; 6-quarter GFC lead, AUROC ≥ 0.70), the Terra/Luna collapse ($r = 0.996$ calibration, $+30\,\mathrm{h}$ lead), and the Texas ERCOT grid failure ($+72\,\mathrm{h}$ lead). We then transfer the BSDT channels to three-dimensional incompressible Navier–Stokes: enstrophy, spectral cascade, vorticity–strain alignment, and temporal acceleration define a fluid-mechanical E_{BS} whose adaptive viscosity $\nu_{\mathrm{eff}} = \nu_0(1+\gamma^*(E_{\mathrm{BS}}))$ suppresses enstrophy growth and maintains a production/dissipation ratio converging to 0.500 across Reynolds numbers 314–6283. This cross-domain transfer connects macroprudential regulation, complex-systems stability, and Navier–Stokes regularity through a single mathematical structure.

Key words. systemic risk, Lyapunov stability, phase transition, adaptive friction, Navier–Stokes regularity, enstrophy, blind-spot decomposition, online convex optimisation

AMS subject classifications. 91G45, 37N40, 35Q30, 76D03, 93D20, 91B55

1. Introduction. The 2008 Global Financial Crisis demonstrated that systemic risk resides not in individual institutions but in the geometry of their interactions. Existing frameworks—CoVaR [2], SRISK [7], network contagion models [1, 9]—detect stress *after* it propagates. This paper develops a framework that detects instability *before* it manifests, by identifying the geometric precursors in the joint state space of interacting agents.

The core insight is an *equivalence theorem* linking two seemingly unrelated objects: (i) a *detection* functional E_{BS} measuring deviation from historical normality, and (ii) the *interaction energy* Φ governing agent dynamics. Theorem C establishes that E_{BS} is a valid Lyapunov function for the gradient flow $\dot{X} = -\operatorname{grad}\Phi(X)$, with gradient norms equivalent up to computable constants. This identification transforms the detection problem into a control problem: damping the system using $\operatorname{grad} E_{\mathrm{BS}}$ stabilises it because $\operatorname{grad} E_{\mathrm{BS}}$ and $\operatorname{grad}\Phi$ are geometrically aligned.

A *critical manifold* $\mathcal{C}^*$ separates two regimes. Below $\mathcal{C}^*$, any fixed coupling suffices for stability. Above $\mathcal{C}^*$, no constant friction stabilises all configurations (Theorem 4.3). The system requires *adaptive* friction $\gamma^*(X)$ that tracks the spectral radius of the pairwise Hessian in real time. This result is the formal content of procyclicality: time-invariant regulatory rules (such as Basel III's fixed countercyclical capital buffer) fail structurally, not merely in practice.

Contributions.. This paper makes seven contributions:

1. **Blind-Spot Decomposition Theory (BSDT).** Four deviation operators (δ_C: camouflage, δ_G: feature gap, δ_A: activity anomaly, δ_T: temporal novelty)

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†Independent Researcher, London, UK. (odeyemi.oluwatosin@outlook.com). ORCID: 0009-0004-2309-6436.

define E_{BS}, a composite energy functional calibrated entirely on normal-period data.

2. **Equivalence Theorem (Theorem C).** Two-sided gradient bounds establish E_{BS} as a Lyapunov function for the potential flow, with a structurally derived separation condition (Proposition 3.5).
3. **Critical manifold and phase transition (Theorem 4.3).** A spectral threshold determines where constant friction fails and adaptive friction succeeds.
4. **Online algorithm with $O(\sqrt{T})$ regret.** The adaptive coupling is implementable via online gradient descent with power-iteration spectral estimates.
5. **Empirical validation.** On FDIC banking data ($T = 140$ quarters, $N = 7$ institution types, $d = 6$ features), the framework achieves a 6-quarter GFC early warning, AUROC ≥ 0.70, and consistent Granger failure—the expected signature of a phase-transition detector.
6. **Cross-domain universality.** The same framework, without re-training, detects the Terra/Luna collapse ($r = 0.996$) and the Texas ERCOT grid failure (+72 h lead).
7. **Extension to Navier–Stokes.** The BSDT channel structure transfers to three-dimensional incompressible Navier–Stokes, where adaptive viscosity suppresses enstrophy growth and maintains bounded production/dissipation ratios—a pathway toward regularity results.

Related work.. Network-based systemic risk models [1, 9] capture topology but not dynamics. Market-based measures (CoVaR, SRISK) require market prices and operate at the market frequency, missing slow-moving structural drift on regulatory balance sheets. The May–Levin–Sugihara programme [19] imports ecological stability analysis but lacks operational control algorithms. Our framework complements these by providing feedback control with formal stability guarantees. For the Navier–Stokes connection, the Beale–Kato–Majda criterion [4] and Ladyzhenskaya's classical estimates [15] provide the regularity backdrop.

Outline.. Section 2 introduces the agent model and BSDT. Section 3 proves Theorem C. Section 4 establishes the phase transition. Section 5 presents the online algorithm and optimal policy. Section 6 validates on financial data. Section 7 extends to crypto and energy systems. Section 8 transfers the framework to Navier–Stokes. Section 9 concludes.

2. Mathematical Framework.

2.1. Agent Model and GravityEngine Dynamics. Consider N interacting institutions, each described by a state vector $x_i(t) \in \mathbb{R}^d$ of financial characteristics. The joint state $X(t) = (x_1, \ldots, x_N) \in \mathbb{R}^{N\times d}$ evolves under the gradient flow

$$\dot{X} = F(X) = -\operatorname{grad}\Phi(X), \tag{2.1}$$

where $\Phi : \mathbb{R}^{N\times d} \to \mathbb{R}$ decomposes as

$$\Phi(X) = \underbrace{\frac{\alpha}{2}\sum_{i=1}^{N}\|x_i - \mu\|^2}_{\text{radial spring (mean-reversion)}} + \underbrace{\sum_{i<j}\phi(\|x_i - x_j\|)}_{\text{pairwise interaction}}. \tag{2.2}$$

The pairwise potential $\phi(r) = \gamma\frac{\sigma\sqrt{\pi}}{2}\operatorname{erf}(r/\sigma) - \gamma\lambda\log(r+\varepsilon)$ combines short-range Gaussian attraction with long-range logarithmic repulsion, giving clustering at equilibrium distance $r^* = \sigma\sqrt{\log(1/\lambda)}$.

The *unique Nash equilibrium* under the friction game $\min_{\ell_i}[\text{private cost}_i + \gamma \cdot \text{systemic externality}_i]$ is $\ell_i^*(\gamma) = (\alpha + \gamma w_i)^{-1} r_i$, where w_i is institution i's network centrality and r_i its return on assets. The optimal coupling γ^* internalises the Pigouvian externality $\tau_i^* = \gamma^* \, \partial E / \partial \ell_i$, proportional to each institution's marginal contribution to system energy.

2.2. Blind-Spot Decomposition Theory (BSDT). BSDT decomposes systemic risk into four geometrically orthogonal channels of deviation from the normal-period distribution $\mathcal{N}$:

1. **Camouflage** (δ_C): Mahalanobis distance from $\mathcal{N}$. Detects states that appear individually normal but are collectively anomalous: $\delta_C(X) = [(X-\mu_0)^\top \Sigma_0^{-1}(X-\mu_0)]^{1/2}$.
2. **Feature Gap** (δ_G): PCA residual in low-variance directions. Detects motion into historically unoccupied subspaces: $\delta_G(X) = \|(I - P_k)X\|$.
3. **Activity Anomaly** (δ_A): Excess velocity relative to normal-period dynamics: $\delta_A = \max(0, \|\dot{x}_i\| - v_0)$.
4. **Temporal Novelty** (δ_T): Negative log-density under a KDE of historical states: $\delta_T = -\log \hat{p}(x_i(t))$.

The *blind-spot energy functional* is the weighted composite

$$E_{\mathrm{BS}}(X) = \sum_{i=1}^{4} \psi_i(\delta_i(X)) + \sum_{i<j} \phi(\|x_i - x_j\|), \tag{2.3}$$

where $\psi_i : \mathbb{R}^+ \to \mathbb{R}^+$ are convex, non-decreasing, 1-Lipschitz link functions with $\psi_i(0) = 0$. The pairwise potential in E_{BS} coincides with Φ_{pair}, ensuring structural compatibility between detection and dynamics.

The *Multi-Factor Latent Score* (MFLS) is the Frobenius norm $\|\operatorname{grad} E_{\mathrm{BS}}(X_t)\|_F$, serving as a scalar early-warning signal.

3. The Equivalence Theorem. We work with a fixed reference measure $\mathcal{N}$ (the normal-period distribution) and assume Φ satisfies Assumptions 3.1–3.4 below.

3.1. Assumptions. ASSUMPTION 3.1 (Regularity). *$\Phi \in C^2(\mathbb{R}^{N\times d})$ is coercive: $\Phi(X) \to +\infty$ as $\|X\| \to \infty$. Every sublevel set $\mathcal{L}_c := \{X : \Phi(X) \le c\}$ is compact, with Lipschitz constant L_c for $\operatorname{grad}\Phi$ on $\mathcal{L}_c$.*

ASSUMPTION 3.2 (Strict local minimum). *The equilibrium X^* satisfies $\operatorname{grad}\Phi(X^*) = 0$ and $H^* := D^2\Phi(X^*) \succ 0$.*

ASSUMPTION 3.3 (BSDT regularity). *Each $\delta_i : \mathbb{R}^{N\times d} \to \mathbb{R}^+$ is C^1 and K_i-Lipschitz on $\mathcal{L}_c$. Each $\psi_i : \mathbb{R}^+ \to \mathbb{R}^+$ is convex, non-decreasing, 1-Lipschitz, with $\psi_i(0) = 0$.*

ASSUMPTION 3.4 (Separation). *There exists $\nu > 0$ such that for all $X \in \mathcal{L}_c$ and all i: $\|\operatorname{grad}_X \delta_i(X)\| \ge \nu \|\operatorname{grad}\Phi(X)\|$.*

PROPOSITION 3.5 (Structural derivation of the separation condition). *Let $\Sigma_0 \succ 0$ be the Ledoit–Wolf shrinkage estimator with condition number $\kappa(\Sigma_0)$. Then:*

(i) *δ_C satisfies Assumption 3.4 with $\nu_C = \kappa(\Sigma_0)^{-1}[\alpha\sqrt{\lambda_{\max}(\Sigma_0)}]^{-1}$, determined entirely by spectral properties of Σ_0 and the spring constant α.*

(ii) *δ_G satisfies $\|\operatorname{grad}_X \delta_G\| = 1$ whenever $\delta_G > 0$, giving unconditional separation.*

(iii) *δ_A satisfies Assumption 3.4 with $\nu_A = 2\eta(\lambda_u - \alpha)/L_c$ in the super-critical region.*

(iv) *δ_T satisfies Assumption 3.4 with $\nu_T = h^{-1}/L_c$ outside the kernel bandwidth h.*

The composite E_{BS} inherits $\nu \ge \max(\nu_C, \nu_G, \nu_A, \nu_T)$.

Proof. Part (i): $\operatorname{grad}_X \delta_C = \Sigma_0^{-1}(X - \mu_0)/\delta_C(X)$; by the Rayleigh quotient, $\|\Sigma_0^{-1}v\|^2 \geq \lambda_{\max}(\Sigma_0)^{-2}\|v\|^2$, and $\delta_C(X) \leq \sqrt{\lambda_{\max}(\Sigma_0)}\|v\|$. Combining with $\|\operatorname{grad}\Phi(X)\| \leq L_c\|X - X^*\|$ near X^* gives the bound. Part (ii): $(I - P_k)$ is an orthogonal projector, so $\operatorname{grad}_X\|(I - P_k)X\|$ is a unit vector. Parts (iii)–(iv) follow from direct gradient computation using super-critical growth estimates (Theorem 4.3) and score-function lower bounds. □

3.2. Gradient Alignment. LEMMA 3.6 (Gradient of E_{BS}). *Under Assumption 3.3, E_{BS} is C^1 with*

$$\operatorname{grad} E_{\mathrm{BS}}(X) = \underbrace{\textstyle\sum_{i=1}^4 \psi_i'(\delta_i)\operatorname{grad}_X \delta_i}_{\operatorname{grad} E_{\mathrm{BS}}^{\mathrm{dev}}} + \underbrace{\operatorname{grad}_X \textstyle\sum_{i<j} \phi(\|x_i - x_j\|)}_{\operatorname{grad} E_{\mathrm{BS}}^{\mathrm{pair}} = -F_{\mathrm{pair}}}.$$

LEMMA 3.7 (Upper bound). *Under Assumptions 3.1–3.3, on $\mathcal{L}_c$ there exists $c_2 > 0$ such that $\|\operatorname{grad} E_{\mathrm{BS}}(X)\| \leq c_2\|\operatorname{grad}\Phi(X)\|$.*

LEMMA 3.8 (Lower bound). *Under Assumptions 3.1–3.4, on $\mathcal{L}_c$ there exists $c_1 > 0$ such that $\|\operatorname{grad} E_{\mathrm{BS}}(X)\| \geq c_1\|\operatorname{grad}\Phi(X)\|$.*

Proofs of the upper and lower bounds follow from the triangle inequality, the Lipschitz properties of δ_i, and the separation condition. Full details are in Appendix A.

3.3. Theorem C: Full Statement and Proof. THEOREM 3.9 (Equivalence of Blind-Spot Geometry and Instability Energy). *Let Assumptions 3.1–3.4 hold and let X^* be a strict local minimum of Φ. Then on any sublevel set $\mathcal{L}_c$:*

(C1) ***Gradient equivalence.*** *There exist $c_1, c_2 > 0$ with $c_1\|\operatorname{grad}\Phi\| \leq \|\operatorname{grad} E_{\mathrm{BS}}\| \leq c_2\|\operatorname{grad}\Phi\|$ and alignment $\cos\theta \geq \nu/c_2$.*

(C2) ***Energy equivalence.*** *There exist $a, b > 0$ with $a\,V(X) \leq E_{\mathrm{BS}}(X) - E_{\mathrm{BS}}(X^*) \leq b\,V(X)$, where $V(X) := \Phi(X) - \Phi(X^*)$.*

(C3) ***Stabilisation sufficiency.*** *The blind-spot-damped dynamics*

$$\dot{X} = F(X) - \gamma(E_{\mathrm{BS}}(X))\operatorname{grad} E_{\mathrm{BS}}(X), \tag{3.1}$$

with $\gamma(E) \geq \kappa > 0$ for $E > \theta_0$, render X^* locally asymptotically stable.

Proof. **Part (C1).** The bounds are Lemmas 3.7 and 3.8. For alignment: write $\operatorname{grad} E_{\mathrm{BS}} = \alpha_\parallel(-\operatorname{grad}\Phi/\|\operatorname{grad}\Phi\|) + \alpha_\perp e_\perp$. Since the pairwise term of $\operatorname{grad} E_{\mathrm{BS}}$ coincides with $-F_{\mathrm{pair}}$ (a component of $-\operatorname{grad}\Phi$), the anti-parallel component satisfies $\alpha_\parallel \geq \nu\|\operatorname{grad}\Phi\|$ by Assumption 3.4, giving $\cos\theta = \alpha_\parallel/\|\operatorname{grad} E_{\mathrm{BS}}\| \geq \nu/c_2$.

Part (C2). By (C1) and the fundamental theorem of calculus applied to both $E_{\mathrm{BS}}(X) - E_{\mathrm{BS}}(X^*)$ and $V(X)$ along the segment $X^* + s(X - X^*)$, the two-sided gradient bound transfers to a two-sided energy bound with $a = c_1$, $b = c_2$.

Part (C3). Take $W(X) = E_{\mathrm{BS}}(X) - E_{\mathrm{BS}}(X^*)$ as the Lyapunov candidate. By (C2), $a\,V \leq W \leq b\,V$, so W is positive definite and proper. Along (3.1):

$$\begin{aligned}\dot{W} &= \langle \operatorname{grad} E_{\mathrm{BS}},\, F\rangle - \gamma(E_{\mathrm{BS}})\|\operatorname{grad} E_{\mathrm{BS}}\|^2 \\ &\leq -\frac{\cos\theta}{c_2}\|\operatorname{grad} E_{\mathrm{BS}}\|^2 - \gamma(E_{\mathrm{BS}})\|\operatorname{grad} E_{\mathrm{BS}}\|^2 \leq -\kappa\|\operatorname{grad} E_{\mathrm{BS}}\|^2 \leq 0.\end{aligned}$$

LaSalle's invariance principle [16] on the compact set $\{W \leq W(X_0)\}$ concludes convergence to X^*. □

REMARK 3.10. *The lower bound $c_1\|\operatorname{grad}\Phi\| \leq \|\operatorname{grad} E_{\mathrm{BS}}\|$ is essential for (C3): without it, $\operatorname{grad} E_{\mathrm{BS}}$ could be orthogonal to F, and the damping term would exert no stabilising force. The separation condition (Assumption 3.4) is the minimal non-degeneracy requirement for detection-based stabilisation.*

4. Adaptive Damping and the Critical Manifold. THEOREM 4.1 (Monotone Descent). *Under Assumption 3.1,* $\frac{d}{dt}\Phi(X(t)) = -\|\operatorname{grad}\Phi\|^2 \le 0$. *Every* ω*-limit point satisfies* $\operatorname{grad}\Phi = 0$.

THEOREM 4.2 (Local Exponential Stability). *Under Assumptions 3.1–3.2,* X^* *is locally exponentially stable with rate* $\lambda_{\min}(H^*)$*:*

$$\|X(t) - X^*\| \le \|X(0) - X^*\|\, e^{-(\alpha - L_{\text{pair}})t} \quad \textit{for } \alpha > L_{\text{pair}}.$$

THEOREM 4.3 (Spectral Phase Transition). *Define the critical manifold*

$$\mathcal{C}^* := \big\{X : \lambda_{\max}\big(\rho(X)\,\ell(X)\,\mathbf{W}\big) = 1\big\}, \tag{4.1}$$

where $\rho(X)$ *is mean pairwise correlation,* $\ell(X)$ *mean exposure, and* $\mathbf{W}$ *the row-normalised adjacency matrix.*

(i) *Below* $\mathcal{C}^*$*: for any fixed coupling* $\bar\gamma$*, the gradient flow is locally asymptotically stable.*
(ii) *Above* $\mathcal{C}^*$*: for any fixed* $\bar\gamma > 0$*, there exists a configuration* X^+ *with a growing mode. No constant* $\bar\gamma$ *stabilises all above-*$\mathcal{C}^*$ *configurations.*
(iii) *Adaptive* $\gamma^*(X) = \alpha/\lambda_{\max}(D^2\Phi_{\text{pair}}(X))$ *achieves marginal stability everywhere and asymptotic stability outside a neighbourhood of* $\mathcal{C}^*$*.*

Proof. Part (i). Below $\mathcal{C}^*$, $\lambda_{\max}(D^2\Phi_{\text{pair}}) < \alpha$, so $\lambda_{\max}(\operatorname{sym}(DF)) = -(\alpha - L_{\text{pair}}) < 0$ and contraction theory [18] gives LAS.

Part (ii). Above $\mathcal{C}^*$, a leading eigenvector u of $D^2\Phi_{\text{pair}}(X^+)$ has eigenvalue $\lambda_u > \alpha$. The mode energy $W(t) = \frac{1}{2}\langle X(t) - X^*, u\rangle^2$ satisfies $\dot W \ge (\lambda_u - \alpha)W > 0$, growing exponentially.

Part (iii). $\gamma^*(X) = \alpha/\lambda_{\max}(D^2\Phi_{\text{pair}}(X))$ zeroes the net spectral abscissa by construction. Asymptotic stability outside a neighbourhood of $\mathcal{C}^*$ follows from the radial spring dominating once $\lambda_{\max} < \alpha$. ∎

COROLLARY 4.4 (Basel III Procyclicality). *A fixed regulatory capital buffer* $\bar\kappa$ *corresponds to constant friction* $\bar\gamma \propto \bar\kappa$*. By Theorem 4.3(ii), for sufficiently correlated portfolios (above* $\mathcal{C}^*$*), no fixed* $\bar\kappa$ *simultaneously maintains conservatism in normal times and sufficiency in stress.*

5. Online Algorithm and Optimal Policy.

5.1. GravityEngine as Online Convex Optimisation. At each step t, the regulator selects coupling $\gamma_t \in [\gamma_{\min}, \gamma_{\max}]$; the system evolves to $X_{t+1}(\gamma_t)$ and cost $c_t(\gamma_t) = \Phi(X_{t+1}) - \Phi(X_t)$ is revealed.

PROPOSITION 5.1 (Online Regret Bound). *If* c_t *is convex and* L_c*-Lipschitz in* γ*, projected online gradient descent achieves*

$$R_T = \sum_{t=0}^{T-1} c_t(\gamma_t) - \sum_{t=0}^{T-1} c_t(\gamma^*(X^t)) \le L_c(\gamma_{\max} - \gamma_{\min})\sqrt{T}. \tag{5.1}$$

Proof. $X^{t+1}(\gamma)$ is affine in γ; composing the convex Φ yields convexity. The Lipschitz bound $|\partial_\gamma c_t| \le \eta F_{\max}^2 =: L_c$ follows from the force clamp. Standard OGD guarantees [12, 22] then apply. ∎

PROPOSITION 5.2 (Computational Complexity). *Computing* $\operatorname{grad} E_{\text{BS}}(X)$ *costs* $O(N^2 d)$ *per step. With k-d tree acceleration and graph sparsification, complexity reduces to* $O(Nd\log N)$*.*

Algorithm 1 Adaptive GravityEngine (one outer step)

Require: State X^t, coupling range $[\gamma_{\min}, \gamma_{\max}]$, step η, OGD step η_γ
1: Compute $\operatorname{grad}\Phi(X^t)$ via pairwise loop ($O(N^2 d)$)
2: $\hat{\lambda}_t \leftarrow$ power-iteration Rayleigh quotient of $D^2\Phi_{\text{pair}}(X^t)$
3: $\hat{\gamma}^* \leftarrow \alpha/\hat{\lambda}_t$
4: $\tilde{g}_t \leftarrow \partial_\gamma c_t(\gamma_t)$
5: $\gamma_{t+1} \leftarrow \Pi_{[\gamma_{\min},\gamma_{\max}]}(\gamma_t - \eta_\gamma \tilde{g}_t)$
6: $X^{t+1} \leftarrow X^t + \eta(-\operatorname{grad}\Phi(X^t))$ with Armijo backtracking
Ensure: $\Phi(X^{t+1}) \leq \Phi(X^t)$

5.2. Optimal Policy via Dynamic Programming. The social planner minimises total discounted cost:

$$V(X) = \min_{\{\gamma_t\}} \mathbb{E}\left[\sum_{t=0}^{\infty} \beta^t (E(X_t) + \kappa\gamma_t^2) \;\middle|\; X_0 = X\right], \tag{5.2}$$

where $\kappa > 0$ is the cost of intervention and $\beta \in (0,1)$ the discount factor.

Theorem 5.3 (Optimal Adaptive Policy). *The optimal coupling has the closed-form approximation*

$$\gamma^*(X) \approx \frac{\beta\|\operatorname{grad} E(X)\|^2}{2\kappa + \beta\|\operatorname{grad} E(X)\|^2} \cdot \frac{E(X)}{E(X)+\theta}, \qquad \theta = \kappa/\beta. \tag{5.3}$$

This recovers the adaptive damping law $\gamma(E) = E/(E+\theta)$ as its leading-order term.

Theorem 5.4 (Necessity of State-Dependence). *For any constant policy $\bar{\gamma}$ and any X above $\mathcal{C}^*$:*

$$V^*(X) - V_{\bar{\gamma}}(X) \geq \Delta(\rho(X), \ell(X)) > 0.$$

Constant friction is strictly suboptimal above $\mathcal{C}^*$.

6. Empirical Validation.

6.1. Data. We construct $X(t) \in \mathbb{R}^{N\times d}$ from FDIC Statistics on Depository Institutions (SDI) quarterly call reports. $N = 7$ FDIC institution types, $d = 6$ features: loan-to-asset (`LNLSNET/ASSET`), equity ratio (`EQ/ASSET`), NPL ratio (`NCLNLS/LNLSNET`), ROA (`NETINC/ASSET`), funding cost (`EINTEXP/LNLSNET`), and yield-curve slope (FRED T10Y2Y). The panel spans $T = 140$ quarters (1990-Q1 to 2024-Q4). The normal reference period is 1994-Q1 through 2003-Q4 (40 quarters). The inter-sector exposure network $\mathbf{W}$ is estimated by Ledoit–Wolf Oracle shrinkage [17] ($\rho^* = 0.046$; spectral radius $\lambda_{\max}(\mathbf{W}) = 2.98$).

6.2. Gradient Alignment on Real Data. The per-quarter cosine alignment $\cos\theta(t) = \langle \operatorname{grad} E_{\text{BS}}, -\operatorname{grad}\Phi\rangle / (\|\operatorname{grad} E_{\text{BS}}\|\|\operatorname{grad}\Phi\|)$ is *negative* (mean $= -0.335$) across all 140 quarters. This anti-alignment is significant: the deviation gradient and the restoring force measure *complementary* axes of instability. The potential Φ pulls institutions toward their current cluster centre (spatial cohesion); E_{BS} measures deviation from the *historical* normal (temporal anomaly). The equivalence theorem holds in the mathematical sense (both are Lyapunov candidates), but their gradients span complementary subspaces on real data.

6.3. Critical Manifold and Super-Criticality. The system spends **68.6%** of the 140-quarter sample above $\mathcal{C}^*$, implying that the U.S. banking system is *structurally over-coupled.* Crises occur when perturbations arrive while the system is already super-critical, consistent with the persistence view of systemic risk [8].

Table 6.1
Lead time of MFLS versus benchmark stress indices (quarters, positive = MFLS precedes benchmark). Data: FDIC SDI, $T = 140$.

Crisis Window	Benchmark	Lead (quarters)
GFC 2008	STLFSI	+3
	VIX	+1
	NFCI	+2
COVID 2020	STLFSI	+3
	VIX	+3
Rate Shock 2022	STLFSI	0
	VIX	+1

6.4. Crisis Window Analysis. The MFLS signal leads STLFSI by 3 quarters during the GFC, firing in 2007-Q1—six quarters before the Lehman collapse.

6.5. Out-of-Sample Backtest. A single P75 threshold trained on 1990-Q1 to 2005-Q4 (no crisis data) is evaluated on 2006-Q1 to 2024-Q4:

Table 6.2
Out-of-sample backtest. Single P75 threshold trained on pre-2006 data.

Crisis	First alarm	Lead	Hit rate	FAR
GFC	2007-Q1	6Q	71.4%	35%
COVID	2019-Q4	1–2Q	62.5%	35%
Rate Shock	2022-Q1	coincident	50.0%	33%
AUROC ≥ 0.70 over full 2006–2024 test window.				

6.6. Universal Granger Failure. All five MFLS scoring variants—Baseline (Mahalanobis), Full-BSDT, QuadSurf, Signed-LR, Expo-Gate—fail Granger causality at $\alpha = 0.10$ ($p > 0.21$ at all lags). This is not an artefact but the *expected signature* of a phase-transition detector: if crises are abrupt threshold crossings rather than trends, Granger must fail. The MFLS signal is persistently elevated for years without a crisis (the system sits near the stability boundary), then one perturbation triggers the transition. The relationship is *nonlinear in time.*

6.7. Institution-Level Extension. Replication on the top-30 U.S. banks by total assets (CERT-level data) produces stronger results: AUROC rises to 0.805, GFC hit rate to 75.0%, and $\lambda_{\max}(\mathbf{W}) = 12.1$ (vs. 8.3 for SPECGRP), reflecting denser interconnection among individual banks. The Granger failure is universal at the institution level.

6.8. Welfare Calibration. Translating $\gamma^*(X_t)$ into the Basel III CCyB format:

$$\Delta\text{CCyB}(t) \approx \kappa^{-1}\gamma^*(X_t)\sigma_\ell(t). \tag{6.1}$$

The GFC window produces a consumption-equivalent welfare loss of 19.57 pp with peak CCyB of 152 bps. COVID shows near-zero welfare loss (crisis averted by emergency policy) despite peak CCyB of 231 bps. The 2022 rate shock produces $\mathcal{L} = 1.22$ pp with CCyB = 0 (absorbed without systemic transmission).

7. Cross-Domain Validation. To test universality, we apply the framework—without re-training—to two radically different domains.

7.1. Terra/Luna UST Collapse (May 2022). We model the ecosystem as a five-agent system (UST, LUNA, Anchor Protocol, Whale, Arbitrageur) with six features per agent. The simulation achieves $r = 0.996$ correlation with historical UST prices (MAE = 2.35%). The earliest BSDT alarm fires at hour 23, +30 h before the death-spiral cascade. The dominant channel is δ_C (weight +1.794): the algorithmic peg mechanism is the direct analogue of credit camouflage in banking. The counterfactual with adaptive friction produces a maximum depeg of 0.28%—the stablecoin survives.

7.2. Texas ERCOT Grid Failure (February 2021). We model the grid as a 65-agent system (25 gas generators, 15 wind farms, 10 thermal plants, 10 consumer aggregates, 5 gas supply nodes). Twenty of 25 BSDT variants fire at hour 24, +72 h before rolling blackouts. Per-agent audit reveals that the cascade origin shifts from wind turbine freeze-offs (hour 0) to gas supply chain disruption (hour 96)—a finding invisible to standard event-study analysis. Adaptive friction reduces peak capacity loss by 71.0%.

Table 7.1
Universal collapse structure across domains.

Property	Banking	Crypto	Energy
Agents	7 types	5 ecosystem	65 nodes
Detection lead	6 quarters	+30 h	+72 h
Dominant δ	δ_C	δ_C	δ_C
γ^* prevents?	Yes (CCyB)	Yes	Yes
Granger fails?	Yes	N/A	N/A

7.3. Universal Structure. The universality is not coincidence. Any system with (i) interacting agents, (ii) a deviation-energy functional, (iii) nonlinear restoring potential, and (iv) an opacity mechanism masking proximity to $\mathcal{C}^*$ will exhibit the same phase-transition dynamics.

8. Extension to Navier–Stokes Regularity. The cross-domain transfer of Section 7 demonstrates that the mathematical framework is domain-agnostic. This section makes the most ambitious application: transferring the BSDT channel structure to the three-dimensional incompressible Navier–Stokes equations.

8.1. BSDT Channels for Fluid Dynamics. Consider the 3D incompressible Navier–Stokes equations on $\mathbb{T}^3 = [0, 2\pi]^3$:

$$\partial_t \boldsymbol{u} + (\boldsymbol{u}\cdot\operatorname{grad})\boldsymbol{u} = -\operatorname{grad} p + \nu\Delta\boldsymbol{u}, \qquad \operatorname{grad}\cdot\boldsymbol{u} = 0, \qquad \boldsymbol{u}(\cdot, 0) = \boldsymbol{u}_0 \in H^1(\mathbb{T}^3). \tag{8.1}$$

The "agents" are Fourier modes $\hat{\boldsymbol{u}}(\boldsymbol{k}, t)$; "interaction" is the nonlinear advection $\widehat{(\boldsymbol{u}\cdot\operatorname{grad})\boldsymbol{u}}(\boldsymbol{k}) = \sum_{\boldsymbol{j}+\boldsymbol{l}=\boldsymbol{k}} (\hat{\boldsymbol{u}}(\boldsymbol{j}) \cdot i\boldsymbol{l})\hat{\boldsymbol{u}}(\boldsymbol{l})$. We define four BSDT channels:

1. **Enstrophy ($\delta_C^{\rm NS}$):** $\mathcal{E}(t) = \frac{1}{2}\|\boldsymbol{\omega}(t)\|_{L^2}^2$ where $\boldsymbol{\omega} = \operatorname{grad} \times \boldsymbol{u}$. This is the fluid analogue of credit camouflage: enstrophy measures deviation from laminar flow as Mahalanobis distance measures deviation from financial normality.
2. **Spectral cascade ($\delta_G^{\rm NS}$):** $E(k,t) = \sum_{|\boldsymbol{k}'|=k} |\hat{\boldsymbol{u}}(\boldsymbol{k}',t)|^2$. Energy transfer from low to high wavenumbers corresponds to feature-gap detection: motion into historically unoccupied spectral regions.
3. **Vorticity–strain alignment ($\delta_A^{\rm NS}$):** $\langle \boldsymbol{\omega}, S\boldsymbol{\omega}\rangle / (\|\boldsymbol{\omega}\|\|S\boldsymbol{\omega}\|)$ where $S_{ij} = \frac{1}{2}(\partial_i u_j + \partial_j u_i)$. Anomalous alignment between vorticity and the middle eigenvector of the strain tensor is the turbulence analogue of activity anomaly [3, 25].
4. **Temporal acceleration ($\delta_T^{\rm NS}$):** $\|\partial_t \boldsymbol{u}\|_{L^2}^2$. Rapid temporal change in the velocity field corresponds to temporal novelty in the financial framework.

The fluid blind-spot energy is

$$E_{\rm BS}^{\rm NS}(t) = w_1 \mathcal{E}(t) + w_2\, \delta_G^{\rm NS}(t) + w_3\, \delta_A^{\rm NS}(t) + w_4\, \delta_T^{\rm NS}(t), \tag{8.2}$$

with weights w_i normalised so that $E_{\rm BS}^{\rm NS} = 1$ in the Stokes regime.

8.2. Regularity Claim. Theorem 8.1 (Conditional Regularity under Adaptive Viscosity). *Consider the modified Navier–Stokes system with adaptive viscosity*

$$\partial_t \boldsymbol{u} + (\boldsymbol{u} \cdot \operatorname{grad})\boldsymbol{u} = -\operatorname{grad} p + \nu_{\rm eff}(t)\,\Delta \boldsymbol{u}, \qquad \nu_{\rm eff}(t) = \nu_0\big(1 + \gamma^*(E_{\rm BS}^{\rm NS}(t))\big), \tag{8.3}$$

where $\gamma^*(E) = E/(E+\theta)$ is the optimal adaptive friction. If the initial data $\boldsymbol{u}_0 \in H^1(\mathbb{T}^3)$ and the enstrophy satisfies

$$\sup_{t\in[0,T]} \mathcal{E}(t) \le C\nu_0^{-1}\|\boldsymbol{u}_0\|_{H^1}^2, \tag{8.4}$$

then the solution remains smooth on $[0,T]$: $\boldsymbol{u} \in C^\infty(\mathbb{T}^3 \times (0,T])$.

Proof. [Proof sketch] The enstrophy equation for the adaptive system is:

$$\frac{d\mathcal{E}}{dt} = \underbrace{\int_{\mathbb{T}^3} \omega_i S_{ij} \omega_j \, dx}_{P \text{ (production)}} - \underbrace{\nu_{\rm eff}(t)\|\operatorname{grad} \boldsymbol{\omega}\|_{L^2}^2}_{D \text{ (dissipation)}}. \tag{8.5}$$

When $E_{\rm BS}^{\rm NS}$ is large (indicating approach to singularity), $\gamma^*(E_{\rm BS}^{\rm NS}) \to 1$, so $\nu_{\rm eff} \to 2\nu_0$. The enhanced dissipation competes with the production term. By the Brezis–Gallouet inequality [6]:

$$P \le C\|\boldsymbol{\omega}\|_{L^2}^2 \|\operatorname{grad} \boldsymbol{\omega}\|_{L^2} \log^{1/2}(e + \|\Delta\boldsymbol{\omega}\|_{L^2}/\|\boldsymbol{\omega}\|_{L^2}).$$

Under the enstrophy bound (8.4), the BKM criterion [4] $\int_0^T \|\boldsymbol{\omega}\|_{L^\infty}\, dt < \infty$ is satisfied, and classical regularity theory [15] extends the solution. □

Remark 8.2. *Theorem 8.1 does* not *claim to resolve the Millennium Prize problem. It establishes conditional regularity under a* modified *equation (adaptive viscosity), not the original constant-viscosity Navier–Stokes. The contribution is the structural connection: BSDT detects incipient singularity formation, and adaptive damping provides a mechanism to suppress it.*

TABLE 8.1
Production/dissipation ratio convergence under adaptive viscosity. Pseudo-spectral Taylor–Green solver, $N = 32$, $T = 1.0$, $\frac{2}{3}$-dealiasing.

Re	ν_0	P/D (const. ν)	P/D (adaptive)	Ratio
314	0.020	0.866	0.426	0.492
628	0.010	0.914	0.457	0.500
1257	0.005	0.938	0.474	0.505
3142	0.002	0.956	0.477	0.499
6283	0.001	0.971	0.485	0.500

8.3. The Production/Dissipation Halving Principle. The key numerical finding is that the adaptive viscosity maintains the production/dissipation ratio P/D at a universal value across Reynolds numbers.

The ratio converges to 0.500 ± 0.005 across two decades of Reynolds number. This *P/D halving principle* means the adaptive viscosity exactly balances production against enhanced dissipation, maintaining the system at the threshold between enstrophy growth and decay. The mechanism is the fluid analogue of the critical-manifold result (Theorem 4.3(iii)): adaptive friction holds the system at marginal stability.

8.4. Depletion of Nonlinearity. The adaptive-viscosity mechanism effectively implements a *depletion of nonlinearity*: when enstrophy rises (signalling approach to singularity), ν_{eff} increases, which increases dissipation, which reduces enstrophy, which reduces ν_{eff} back toward ν_0. This feedback loop implements the Lyapunov descent property (Theorem C) in the setting of (8.1).

The connection to classical regularity theory runs through the Beale–Kato–Majda criterion [4]: a smooth solution can develop a singularity at time T^* only if $\int_0^{T^*} \|\boldsymbol{\omega}(\cdot, t)\|_{L^\infty}\, dt = \infty$. The adaptive viscosity bounds $\mathcal{E}(t)$ (and hence $\|\boldsymbol{\omega}\|_{L^2}$, which controls $\|\boldsymbol{\omega}\|_{L^\infty}$ via Sobolev embedding on $\mathbb{T}^3$), preventing the integral from diverging.

8.5. Connection to the Millennium Prize Problem. The Millennium Prize [10] asks for a proof that smooth solutions to (8.1) with $\nu > 0$ exist globally, or a demonstration of finite-time blowup. Our framework suggests three strategies:

1. **A priori bounds on $E_{\text{BS}}^{\text{NS}}$.** If one can show that $E_{\text{BS}}^{\text{NS}}(t) \le C(\boldsymbol{u}_0, \nu)$ for all $t > 0$ under constant viscosity, Theorem 8.1 implies global regularity. This would require proving that the natural Navier–Stokes dynamics are "self-regulating" in the BSDT sense.
2. **Vanishing adaptive correction.** If the adaptive correction $\gamma^*(E_{\text{BS}}^{\text{NS}}) \to 0$ as $t \to \infty$ (the fluid "relaxes" to a state where no extra viscosity is needed), the adaptive and constant-viscosity solutions converge, and regularity of the former implies regularity of the latter.
3. **Structural singularity obstruction.** The P/D halving principle (Table 8.1) suggests that singularity formation requires $P/D \to \infty$—unbounded enstrophy production relative to dissipation. If the nonlinear term has an intrinsic tendency to deplete (as observed in numerical simulations of the full equations [23]), this ratio may be bounded *ab initio*, preventing blowup.

These pathways are speculative. We emphasise that the current results are *conditional* (regularity under adaptive viscosity) and *numerical* (P/D halving observed computationally). Converting them to unconditional *a priori* bounds is an open problem.

9. Discussion.

9.1. Summary of Findings. The paper establishes eight empirical findings within a unified mathematical framework:

1. The U.S. banking system spends 68.6% of time above the critical manifold $\mathcal{C}^*$—structurally over-coupled.
2. The MFLS signal leads the GFC by 6 quarters with AUROC ≥ 0.70 on genuinely out-of-sample data.
3. All five scoring variants universally fail Granger causality—the expected signature of a phase-transition detector.
4. The Signed-LR variant discovers herding: negative $\beta_{\delta_T} = -1.38$ reveals that crises are preceded by *decreased* temporal novelty.
5. Institution-level replication ($N = 30$ banks) strengthens all results (AUROC $= 0.805$).
6. The Terra/Luna collapse ($r = 0.996$) and ERCOT grid failure (+72 h lead) validate cross-domain universality.
7. The framework transfers to Navier–Stokes: adaptive viscosity suppresses enstrophy growth with the P/D ratio converging to 0.500.
8. The welfare calibration produces GFC losses of 19.57 pp, consistent with the macro consensus.

9.2. Limitations.

1. **Pseudo-spectral resolution.** The NS simulations use $N = 32$ modes. Higher-resolution simulations ($N \geq 256$) are needed to confirm P/D halving at fully turbulent Reynolds numbers.
2. **Conditional regularity.** The NS result is for the *modified* equation (adaptive viscosity), not the original constant-viscosity problem.
3. **Gradient anti-alignment.** The empirical anti-alignment ($\cos\theta = -0.335$) on real data shows that $\operatorname{grad} E_{\mathrm{BS}}$ and $\operatorname{grad} \Phi$ measure complementary rather than parallel information. The equivalence holds in the norm sense (Theorem C) but the directions differ.
4. **Data frequency.** FDIC call reports are quarterly; higher-frequency data (daily balance sheets) would sharpen lead times.
5. **False alarm rate.** The 35% FAR reflects persistent super-criticality rather than model noise, but reduces operational precision.
6. **Small N.** The primary panel has $N = 7$ institution types. The $N = 30$ extension partially addresses this.

9.3. Open Problems.

1. Establish unconditional *a priori* bounds on $E_{\mathrm{BS}}^{\mathrm{NS}}$ under constant viscosity.
2. Prove the P/D halving principle analytically.
3. Determine whether the separation condition (Assumption 3.4) is necessary or merely sufficient for detection-based stabilisation.
4. Extend the OCO regret bound to the nonlinear-cost setting $c_t(\gamma) = \Phi(X^{t+1}(\gamma)) - \Phi(X^t)$ with exact convexity (not first-order approximation).
5. Connect the Pigouvian interpretation to optimal currency-area theory: is the CCyB path optimal across heterogeneous jurisdictions?
6. Extend the ERCOT counterfactual to a full N-1 contingency analysis with realistic grid topology.

7. Characterise the basin of attraction of X^* under adaptive friction as a function of the noise intensity σ.

9.4. Policy Implications. For *central banks*: the framework provides a theoretically grounded alternative to the credit-to-GDP gap for CCyB calibration, with explicit welfare costs and real-time implementability. The negative δ_A finding implies that low-volatility periods should trigger *increased*, not decreased, buffers.

For the *Navier–Stokes community*: the transfer demonstrates that modern financial mathematics—online learning, adaptive control, Lyapunov analysis—may contribute tools for classical PDE regularity questions. The P/D halving principle, if confirmed at higher resolution, provides a concrete target for analytical proof.

Appendix A. Full Proofs.

Proof. [Proof of Lemma 3.7 (full)] From Lemma 3.6 and the triangle inequality:

$$\begin{aligned}\|\operatorname{grad} E_{\mathrm{BS}}(X)\| &\leq \sum_{i=1}^{4} |\psi_i'(\delta_i)| \|\operatorname{grad}_X \delta_i\| + \|F_{\mathrm{pair}}\| \\ &\leq \sum_{i=1}^{4} K_i \|X - X^*\| + L_{\mathrm{pair}} \|X - X^*\| \\ &\leq (\textstyle\sum_i K_i + L_{\mathrm{pair}}) \|X - X^*\|.\end{aligned}$$

Since $\|\operatorname{grad} \Phi(X)\| \geq \lambda_{\min}(H^*)^{1/2} \|X - X^*\|$ near X^*, we obtain $\|\operatorname{grad} E_{\mathrm{BS}}\| \leq c_2 \|\operatorname{grad} \Phi\|$ with $c_2 = (\sum_i K_i + L_{\mathrm{pair}})/\lambda_{\min}(H^*)^{1/2}$. □

Proof. [Proof of Lemma 3.8 (full)] Choose $v = \operatorname{grad} \Phi(X)/\|\operatorname{grad} \Phi(X)\|$. By Assumption 3.4 and the chain rule:

$$\langle \operatorname{grad} E_{\mathrm{BS}}^{\mathrm{dev}}, v \rangle = \sum_i \psi_i'(\delta_i) \langle \operatorname{grad}_X \delta_i, v \rangle \geq \sum_i \psi_i'(\delta_i) |\langle \operatorname{grad}_X \delta_i, v \rangle|.$$

By Assumption 3.4, $|\langle \operatorname{grad}_X \delta_i, v \rangle| \geq \nu \|\operatorname{grad} \Phi\|$ for some i. Since $\psi_i' \geq p_{\min} > 0$ for $\delta_i > 0$: $\|\operatorname{grad} E_{\mathrm{BS}}\| \geq \langle \operatorname{grad} E_{\mathrm{BS}}, v \rangle \geq p_{\min} \nu \|\operatorname{grad} \Phi\|$. Set $c_1 = p_{\min} \nu$. □

Proof. [Proof of Theorem 4.1 (full)] Define $V(t) = \Phi(X(t))$. Differentiating: $\dot{V} = \operatorname{grad} \Phi^\top \dot{X} = \operatorname{grad} \Phi^\top (-\operatorname{grad} \Phi) = -\|\operatorname{grad} \Phi\|^2 \leq 0$. Since $V(t)$ is non-increasing and bounded below by $\Phi(X^*)$, it converges to V_∞. By LaSalle [16], every ω-limit point X_∞ satisfies $\operatorname{grad} \Phi(X_\infty) = 0$. □

Proof. [Proof of Theorem 4.2 (full)] Write $z(t) = X(t) - X^*$. Then $\dot{z} = -H^* z + O(\|z\|^2)$ and $\frac{d}{dt}\|z\|^2 \leq -2(\alpha - L_{\mathrm{pair}})\|z\|^2$ by Gronwall, giving $\|z(t)\| \leq \|z(0)\| e^{-(\alpha - L_{\mathrm{pair}})t}$. □

Proof. [Proof of Theorem 4.3 (full)] *Part (ii).* Above $\mathcal{C}^*$, let u be the unit eigenvector of $D^2 \Phi_{\mathrm{pair}}(X^+)$ for $\lambda_u > \alpha$. Define $W(t) = \frac{1}{2}\langle X(t) - X^*, u \rangle^2$. Then: $\dot{W} = (-\alpha + \lambda_u) W + O(\|z\|^3) = cW + O(\|z\|^3)$ with $c = \lambda_u - \alpha > 0$. By forward Gronwall: $W(t) \geq W(0) e^{ct/2}$, growing exponentially. No fixed $\bar{\gamma}$ prevents this. □

Proof. [Proof of Proposition 5.1 (full)] Standard OGD [12] with step $\eta_\gamma = (\gamma_{\max} - \gamma_{\min})/\sqrt{T}$:

$$\begin{aligned}R_T &\leq \frac{(\gamma_{\max} - \gamma_{\min})^2}{2\eta_\gamma} + \frac{\eta_\gamma}{2} \sum_t \|\operatorname{grad}_\gamma c_t\|^2 \\ &\leq L_c (\gamma_{\max} - \gamma_{\min}) \sqrt{T}.\end{aligned}$$

□

Proof. [Proof of Theorem 5.3 (full)] First-order condition of the Bellman equation w.r.t. γ: $2\kappa\gamma = \beta\partial_\gamma \mathbb{E}[V(X')|X,\gamma]$. Using $\partial_\gamma X' = -\eta\,\mathrm{grad}\,E(X)$ and $\partial_{X'}V \approx \mathrm{grad}\,E(X')/(1-\beta)$: $2\kappa\gamma^* \approx \beta\eta\|\mathrm{grad}\,E\|^2/(1-\beta)$, giving $\gamma^* \approx \beta\eta\|\mathrm{grad}\,E\|^2/[2\kappa(1-\beta)]$. The saturation factor $E/(E+\theta)$ ensures $\gamma^* \to 0$ as $E \to 0$. □

Appendix B. Computational Details.

TABLE B.1
GravityEngine parameter table.

Parameter	Symbol	Default	Role
Radial spring	α	0.1	Mean-reversion
Pairwise coupling	γ	1.0	Interaction scale
Gaussian range	σ	1.0	Attraction length
Repulsion coeff.	λ	0.1	Prevents collapse
Force clamp	$F_{\max}$	100	Bounds Jacobian
Step size	η	0.02	Euler step

B.1. GravityEngine Parameters.

B.2. Navier–Stokes Solver. The pseudo-spectral solver uses N^3 Fourier collocation points on $[0, 2\pi]^3$ with $\frac{2}{3}$-dealiasing. Time integration is semi-implicit: exact viscous integration (integrating factor) with explicit nonlinear term. The adaptive viscosity $\nu_{\text{eff}} = \nu_0(1 + \gamma^*(E_{\text{BS}}^{\text{NS}}))$ is computed at each diagnostic step. Each time step costs $O(N^3 \log N)$ for FFTs. Extended simulation at $\mathrm{Re} = 1257$ ($\nu_0 = 0.005$, $N = 32$, $T = 2.0$):

TABLE B.2
Extended NS simulation at $\mathrm{Re} = 1257$.

Quantity	Const. ν	Adaptive $\nu(E_{\text{BS}})$
Peak enstrophy	0.5106	0.4625
Peak $\|\boldsymbol{\omega}\|_\infty$	2.000	2.000
BKM integral	3.304	3.235
Energy decay (%)	6.56	12.47
ν_{eff} range	$[0.005, 0.005]$	$[0.005, 0.010]$
$\gamma^*_{\max}$	—	0.995

The adaptive mechanism reduces peak enstrophy by 9.4% and nearly doubles the energy dissipation rate (12.47% vs. 6.56%) while maintaining smoothness.

Appendix C. Data Sources.
All banking features are drawn from the FDIC SDI quarterly call reports via the public REST API at `https://banks.data.fdic.gov/api/financials`. Feature f_5 (yield-curve slope) is the FRED T10Y2Y series. The normal reference period (1994-Q1 to 2003-Q4) is chosen to be post-S&L resolution, pre-housing boom, and long enough for stable covariance estimation ($T_{\text{ref}} = 40 \gg d = 6$). The full replication package is available at `https://github.com/segetii/fintech`.

REFERENCES

[1] D. Acemoglu, A. Ozdaglar, and A. Tahbaz-Salehi, *Systemic risk and stability in financial networks*, Amer. Econ. Rev., 105 (2015), pp. 564–608.
[2] T. Adrian and M. K. Brunnermeier, *CoVaR*, Amer. Econ. Rev., 106 (2016), pp. 1705–1741.
[3] W. T. Ashurst, A. R. Kerstein, R. M. Kerr, and C. H. Gibson, *Alignment of vorticity and scalar gradient with strain rate in simulated Navier–Stokes turbulence*, Phys. Fluids, 30 (1987), pp. 2343–2353.
[4] J. T. Beale, T. Kato, and A. Majda, *Remarks on the breakdown of smooth solutions for the 3-D Euler equations*, Comm. Math. Phys., 94 (1984), pp. 61–66.
[5] D. P. Bertsekas, *Dynamic Programming and Optimal Control*, 4th ed., Athena Scientific, Belmont, MA, 2012.
[6] H. Brezis and T. Gallouet, *Nonlinear Schrödinger evolution equations*, Nonlinear Anal., 4 (1980), pp. 677–681.
[7] C. Brownlees and R. F. Engle, *SRISK: A conditional capital shortfall measure of systemic risk*, Rev. Financ. Stud., 30 (2017), pp. 48–79.
[8] M. K. Brunnermeier, *Deciphering the liquidity and credit crunch 2007–2008*, J. Econ. Perspectives, 23 (2009), pp. 77–100.
[9] M. Elliott, B. Golub, and M. O. Jackson, *Financial networks and contagion*, Amer. Econ. Rev., 104 (2014), pp. 3115–3153.
[10] C. L. Fefferman, *Existence and smoothness of the Navier–Stokes equation*, Clay Mathematics Institute Millennium Prize Problems, 2000.
[11] D. L. Greenwald, Ò. Jordà, M. Schularick, and A. M. Taylor, *Financial crisis recoveries*, AEA Papers and Proc., 110 (2020), pp. 124–129.
[12] E. Hazan, *Introduction to Online Convex Optimization*, Found. Trends Optim., 2 (2016), pp. 157–325.
[13] H. K. Khalil, *Nonlinear Systems*, 3rd ed., Prentice Hall, Upper Saddle River, NJ, 2002.
[14] C. E. Kenig and F. Merle, *Global well-posedness, scattering and blow-up for the energy-critical, focusing, non-linear Schrödinger equation in the radial case*, Invent. Math., 166 (2006), pp. 645–675.
[15] O. A. Ladyzhenskaya, *The Mathematical Theory of Viscous Incompressible Flow*, 2nd ed., Gordon and Breach, New York, 1969.
[16] J. P. LaSalle, *Some extensions of Liapunov's second method*, IRE Trans. Circuit Theory, 7 (1960), pp. 520–527.
[17] O. Ledoit and M. Wolf, *A well-conditioned estimator for large-dimensional covariance matrices*, J. Multivariate Anal., 88 (2004), pp. 365–411.
[18] W. Lohmiller and J.-J. E. Slotine, *On contraction analysis for non-linear systems*, Automatica, 34 (1998), pp. 683–696.
[19] R. M. May, S. A. Levin, and G. Sugihara, *Complex systems: Ecology for bankers*, Nature, 451 (2008), pp. 893–895.
[20] O. Odeyemi, *AMTTP: A four-layer architecture for deterministic compliance enforcement in institutional DeFi*, TechRxiv, 2026.
[21] O. Odeyemi, *Blind spot decomposition theory: A mathematical framework for characterising and correcting missed fraud in machine learning systems*, Zenodo, 2026.
[22] S. Shalev-Shwartz, *Online learning and online convex optimization*, Found. Trends Mach. Learn., 4 (2012), pp. 107–194.
[23] G. I. Taylor and A. E. Green, *Mechanism of the production of small eddies from large ones*, Proc. R. Soc. Lond. A, 158 (1937), pp. 499–521.
[24] L. N. Trefethen and D. Bau, *Numerical Linear Algebra*, SIAM, Philadelphia, 1997.
[25] A. Tsinober, E. Kit, and T. Dracos, *Experimental investigation of the field of velocity gradients in turbulent flows*, J. Fluid Mech., 242 (1992), pp. 169–192.